

Two Fundamental Algorithms in Quantum Computation

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June 3, 2026

Deutsch–Jozsa Algorithm

Grover's Algorithm

Setup

- We are given a black box quantum computer known as oracle that implements some function

$$f: \{0, 1\} \rightarrow \{0, 1\}.$$

- We are promised that the function is either **constant** (0 on all inputs or 1 on all inputs) or **balanced** (1 for exactly half of the input domain and 0 for the other half).

a	$f_1(a)$	a	$f_2(a)$	a	$f_3(a)$	a	$f_4(a)$
0	0	0	0	0	1	0	1
1	0	1	1	1	0	1	1

Figure: Four outputs of function f , where the first and last of these are constant and the middle two are balanced.

- Algebraically, f is **constant** if $f(0) \oplus f(1) = 0$, **balanced** if $f(0) \oplus f(1) = 1$.
($\oplus$ = "addition modulo 2".)

Algorithm

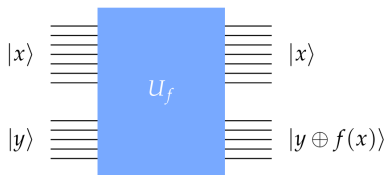


Figure: The action of a unitary query gate U_f on standard basis inputs.

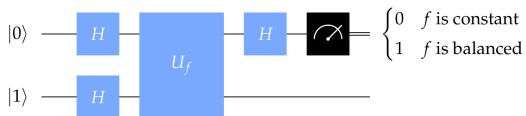
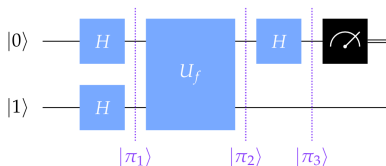


Figure: Deutsch-Jozsa Algorithm.

Algorithm



$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad |\pi_1\rangle = |-\rangle |+\rangle = \frac{1}{2}(|0\rangle - |1\rangle)|0\rangle + \frac{1}{2}(|0\rangle - |1\rangle)|1\rangle$$

$$|\pi_2\rangle = \frac{1}{2}(|0 \oplus f(0)\rangle - |1 \oplus f(0)\rangle) = \begin{cases} (-1)^{f(0)} |-\rangle |+\rangle & \text{if } f(0) \oplus f(1) = 0, \\ (-1)^{f(0)} |-\rangle |-\rangle & \text{if } f(0) \oplus f(1) = 1. \end{cases}$$

$$|\pi_3\rangle = \begin{cases} (-1)^{f(0)} |-\rangle |0\rangle & \text{if } f(0) \oplus f(1) = 0 \\ (-1)^{f(0)} |-\rangle |1\rangle & \text{if } f(0) \oplus f(1) = 1 \end{cases}$$

Setup

- ▶ As input for Grover's Algorithm, suppose we have a function $f: \{0, 1, \dots, N-1\} \rightarrow \{0, 1\}$. We assume that only one index satisfies $f(x) = 1$, and we call this index ω . Our goal is to identify ω . (Imagine a box full of balls, among which one is marked.)
- ▶ We can access f with an oracle in the form of a unitary operator U_ω that acts as follows:

$$\begin{cases} U_\omega |x\rangle = -|x\rangle & \text{for } x = \omega, \text{ that is, } f(x) = 1, \\ U_\omega |x\rangle = |x\rangle & \text{for } x \neq \omega, \text{ that is, } f(x) = 0. \end{cases}$$

This is often written as

$$U_\omega |x\rangle = (-1)^{f(x)} |x\rangle.$$

Computation Advantage

- ▶ Classical search: If you search an unsorted list, you might need up to $O(N)$ queries.
- ▶ Grover's quantum search: You can do it in only queries $O(\sqrt{N})$ — a quadratic speedup.

Algorithm

- ▶ (1) Initialize the system to the uniform superposition over all states

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

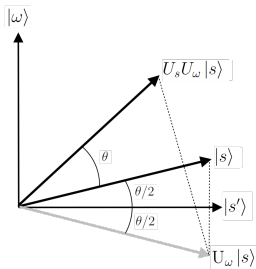
- ▶ (2) Repeatedly perform the following "Grover iteration":
 1. Apply the operator $U_\omega = I - 2|\omega\rangle\langle\omega|$;
 2. Apply the diffusion operator $U_s = 2|s\rangle\langle s| - I$.
- ▶ (3) Measure the resulting quantum state in the computational basis.

Geometric Interpretation

- Consider the plane spanned by $|s\rangle$ and $|\omega\rangle$; equivalently, the plane spanned by $|\omega\rangle$ and perpendicular ket

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle.$$

- The operator U_ω is a reflection at the hyperplane orthogonal to $|\omega\rangle$ for vectors in the plane spanned by $|s'\rangle$ and $|\omega\rangle$, i.e. it acts as a reflection across $|s'\rangle$.
- The operator $U_s = 2|s\rangle\langle s| - I$ is a reflection through $|s\rangle$.



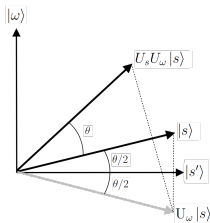
Geometric Interpretation

It is straightforward to check that the operator $U_s U_\omega$ of each Grover iteration step rotates the state vector by an angle of

$$\theta = 2 \arcsin \frac{1}{\sqrt{N}}.$$

So, with enough iterations, one can rotate from the initial state $|s\rangle$ to the desired output state $|\omega\rangle$:

$$\langle s' | s \rangle = \sqrt{\frac{N-1}{N}}.$$



Number of Iterations

- Over k iterations, the state is at angle

$$\frac{2k+1}{2}\theta.$$

We want

$$\frac{2k+1}{2}\theta \approx \frac{\pi}{2},$$

hence

$$k \approx \frac{\pi}{2\theta} - \frac{1}{2}.$$

Number of Iterations

- ▶ When N is sufficiently large, we have

$$\sin \theta = \frac{1}{\sqrt{N}} \Rightarrow \frac{\theta}{2} \approx \frac{1}{\sqrt{N}} \Rightarrow \theta \approx \frac{2}{\sqrt{N}}$$

- ▶ With all above, we conclude that

$$k \approx \frac{\pi}{4} \sqrt{N} - \frac{1}{2},$$

which gives a quadratic speedup.

Backup

The state $|s\rangle$ is the **uniform superposition over all states**:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

Split this sum into: The marked element ($x = \omega$), giving

$$\frac{1}{\sqrt{N}} |\omega\rangle.$$

All unmarked elements ($x \neq \omega$), giving

$$\frac{1}{\sqrt{N}} \sum_{x \neq \omega} |x\rangle.$$

Define

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle.$$

Backup

So we can write

$$|s\rangle = \frac{1}{\sqrt{N}}|\omega\rangle + \sqrt{\frac{N-1}{N}}|s'\rangle.$$

Now check orthogonality:

$$\langle\omega|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} \langle\omega|x\rangle = 0,$$

because $|\omega\rangle$ is not included in the sum, and computational basis states are orthogonal.